Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en Inligtingtegnologie / Lefapha la Boetšenere, Tikologo ya Kago le Theknolotši ya Tshedimošo

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

DEPARTMENT OF INFORMATICS

INFORMATICS 271 MEMO

SEMESTER TEST **DATE: 14 October 2019**

Examiners	: Dr MJ Hattingh : Mrs K Pillay		**Time**	:	120 min
Moderator / External Examiner	: Dr L Weilbach		**Marks**	:	65

Student Number								Surname	Initials

Question / Section	Module outcomes (as in Study Guide)					Marks allocated	Maximum mark
	MO1	MO2	MO3	MO4	MO5		
1				20			20
2				20			20
3							8
4							5
5							6
6							6

Total		65

Instructions

1. This paper consists of 1 section and 6 questions.
2. Answer all the questions – there are no optional questions.
3. This is a closed book test. You may not refer to any printed or electronic material.
4. Please write legibly.
5. Write down your answers on this test paper.

SECTION A – QRMoney Case Study

Read the case study and then answer the questions that follow:

QRMoney is the brainchild of Mr E.N. Trepeneur. It is a system that allows various informal service providers (SP) such as car guards, fruit sellers, arts and craft vendors etc. a way to be active in the digital economy.

The QR system works as follows: A SP buys a QRMoney card (with a QR code on it) from a QRMoney enabled kiosk at Pick n Pay, Checkers, Spar or Dischem stores for R5-00. The sale of this QRMoney Card happens at any till point. On purchase it is not captured on the QRMoney system and forms part of the normal "stock" of the specific store. It is possible for a SP to buy more than one QRMoney Card.

For a SP to make use of his QRMoney card, the QRMoney card will have to be activated on the QRMoney system. To activate the card, the cashier needs to log in on the QRMoney system, using a username and password. S/he will then scan the QRMoney Card after which the system will prompt the Cashier to register the SP by capturing his/her Name, Surname, Mobile Number, and ID Number or Passport Number. Note that the SP's details can be updated at any time. A SP can only be deleted from the system after 5 years if s/he has a balance of R0. As soon as the QRMoney card has been registered on the system, the SP will receive a confirmation SMS.

The SP will then carry the QRMoney card with him/her. When s/he renders a service to a customer and a customer wants to pay him/her (if they don't have cash at hand), the SP presents his/her QRMoney card. The customer will then use Snapscan, Zapper or FNB Pay to pay the SP by accessing the chosen application, scanning the SP's QRMoney Card, and capturing the amount and details required by their application. The transaction is then stored on the database of the QRMoney system, linked to the SP's profile.

When the SP wants to redeem the funds on his/her QRMoney card, s/he can go to any QRMoney enabled kiosk. S/he will ask the cashier to redeem his/her QRMoney by handing his/her QRMoney card to the cashier, who will scan the QRMoney card. The system will retrieve the SP's details and display the amount available in his/her account. The cashier will ask the SP for his ID/Passport document (to verify that the account belongs to the person that redeems the funds) and will then communicate the amount available for withdrawal to the SP and ask him/her how much s/he wants to redeem. The amount available for withdrawal equals the money paid to the SP by his/her clients, less 1% which is paid to the owner of the QRMoney system for the privilege to use of their system. (This percentage is not fixed and could be adjusted by die owner of the QRMoney system at any time). A SP can choose to either redeem the amount as a cash pay-out or as a voucher for the store where they redeem their money. The cashier will input the amount redeemed and select either the 'Cash out' option or the 'Voucher' option. The system will verify that the amount redeemed is less than the amount redeemable, subtract the redeemed amount from the amount redeemable and update the SP's available balance. For the 'Cash out' option the register drawer is opened and the cashier hands over the cash. The SP's profile is updated with the amount redeemed (this is a tally that will allow the SP to track how much money s/he has been paid over time) and the date of redemption. An SMS is sent to the SP confirming the amount redeemed and his/her new balance. For the voucher option, the system will generate a voucher and save the voucher number. The cashier will print the voucher and hand it to the SP.

The SP can also at any point in time approach the cashier at the kiosk and request the cashier the check his/her balance on the system. This is done free of charge. For a printout of his/her account, an amount of R2 is subtracted from his/her account. This is of course only possible if the SP has an available balance of at least R2.

The QRMoney system is managed by the Store Administrator (Store Admin). The Store Admin is responsible to register and update/delete the cashiers' access to the system and is also responsible to set the percentage which has to be paid over to the QRMoney system owner.

The system also allows the stakeholders to generate some reports. This includes:

- For Mr Trepeneur (QRMoney owner): a Monthly Sales Report which indicates the number of system users per kiosk; the number of transactions per user; as well as the amount accumulated through the % subtracted from the SP's income (paid towards the usage of the system).
- For the SP: a monthly SMS with a summary of the transactions recorded for him/her, the withdrawals s/he made, as well as the amount still available for redemption.

Technical Use Case Narrative Extract – UC5.1 Redeem QRMoney

USE CASE NAME:	**Redeem QRMoney**	**USE CASE TYPE**	
USE CASE ID:	**5.1**	**Business Requirements:**	☐
PRIORITY:	High	**System Analysis:**	☐
SOURCE:	QRMoney Case Study	**System Design:**	☑
PRIMARY BUSINESS ACTOR:	Service Provider (SP) - PBA		
PRIMARY THE SYSTEM ACTOR:	Cashier - PSA		
DESCRIPTION:	This use case describes the event where a SP approaches the cashier to redeem his/her money. This can either be done in cash or with a store voucher.		
PRE-CONDITION:	• The Cashier must be logged into the system. • The SP must be activated.		
TRIGGER:	A SP wants to redeem his/her QRMoney and presents his/her QRMoney card to the cashier.		

TYPICAL COURSE OF EVENTS:	**ACTOR ACTION:**	**SYSTEM RESPONSE:**	
		MANUAL ACTION:	**AUTOMATED ACTION:**
	1. A SP wants to redeem his/her QRMoney and presents his/her QRMoney card to cashier.	**2.** The cashier navigates to the home screen, clicks on the Redemption menu item, then clicks on the Redeem QRMoney menu option on the Redemption screen and scans the QRMoney card.	**3.** The system responds by loading the "Redeem QRMoney" screen with the following controls (all controls are enabled unless specified otherwise): Label: • "Service Provider Details:" • SP information section (labels): • Name • Surname • ID/ Passport No • Date Picker corresponding to date of redemption (auto-filled with the current date). • Available Balance • Label: "Amount to redeem" • Text box to enter the amount • Label: "Method to redeem" • "Redeem Cash" and "Redeem Voucher" buttons
		4. The cashier requests the SP to present his/her ID/passport for verification.	
	5. SP presents his/her verification document.	**6.** The cashier verifies that the details on the ID/passport document matches the displayed details. (alt)	
		7. The cashier communicates the SP's available balance to the SP and asks the SP for the amount that s/he wishes to redeem (The	

		available balance is calculated with the following formula: (*(1-percentage)*balance)).*	
	8. SP provides the amount to be redeemed.	**9.** The Cashier captures the amount to be redeemed.	**10.** The system verifies that the amount to be redeemed is less than or equal to the redeemable amount. (alt)
		11. The cashier asks the SP how s/he would like to redeem the amount.	
	12. The SP chooses to redeem the amount as cash. (alt)	**13.** The Cashier clicks the "Redeem Cash" button. (alt)	**14.** The system uses an SQL update query to update the: • SP balance in the SP entity by deducting the redeemed amount from the current amount • Redeem_Transaction table with the transaction date (current system date) • the amount redeemed in the Redeem_Transaction table The cash register drawer is opened.
			15. The system displays the "Cash Notification" message box indicating the SP for whom the transaction was, as well as the amount redeemed and the fact that is was redeemed as cash.
		17. Cashier hands over money to SP.	**16.** System generates the SMS using a cellular provider to notify the SP of the successful transaction and new balance.
ALTERNATE	**Alt step 6:** The details on the SP's ID/passport document does not match the displayed details; the cashier communicates to the SP that he will not be able to complete the transaction and cancels the transaction.		
	Alt Step 10: The amount to be redeemed is more than available amount and the cashier informs the SP of the maximum amount available for redemption. Return to step 7.		
COURSES:	**Alt step 12:** The SP chooses to redeem the amount as a voucher. The cashier clicks the "Redeem Voucher" button as the redemption method. The system uses an SQL update query to update: the SP balance in the SP entity by deducting the redeemed amount from the current amount, the Redeem_Transaction entity with the transaction date (current system date) and the amount redeemed in the Redeem_Transaction table. The system uses an SQL create query to add the voucher number to the Voucher entity. The system displays the "Voucher Notification" message box indicating the SP for whom the transaction was, as well as the amount redeemed and the fact that is was redeemed as a voucher. The system prints out the voucher. The cashier hands over the voucher to the SP.		

Question 1 [20]

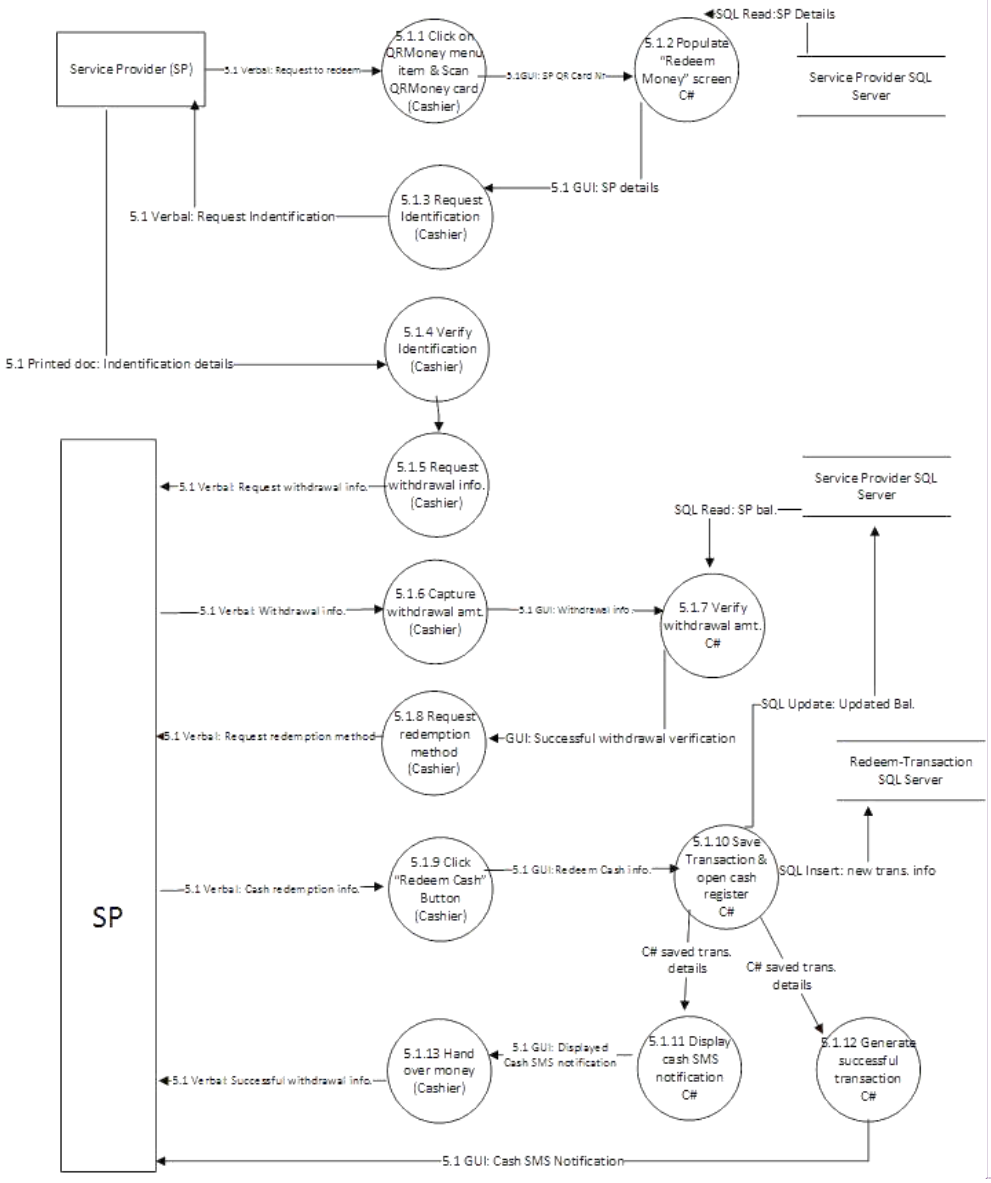
Complete the following **Technical Primitive Level DFD** for use case **5.1 Redeem QRMoney**. The diagram below is complete until step 6 of the given use case narrative.
Only draw the typical course of events.

Processes – 9 (1 mark for each)

Data Stores – 2 (1 mark for each)

Data flows – 9 (1/2 mark for each flow)

UC 5.1 Redeem QR Money Primitive Level DFD

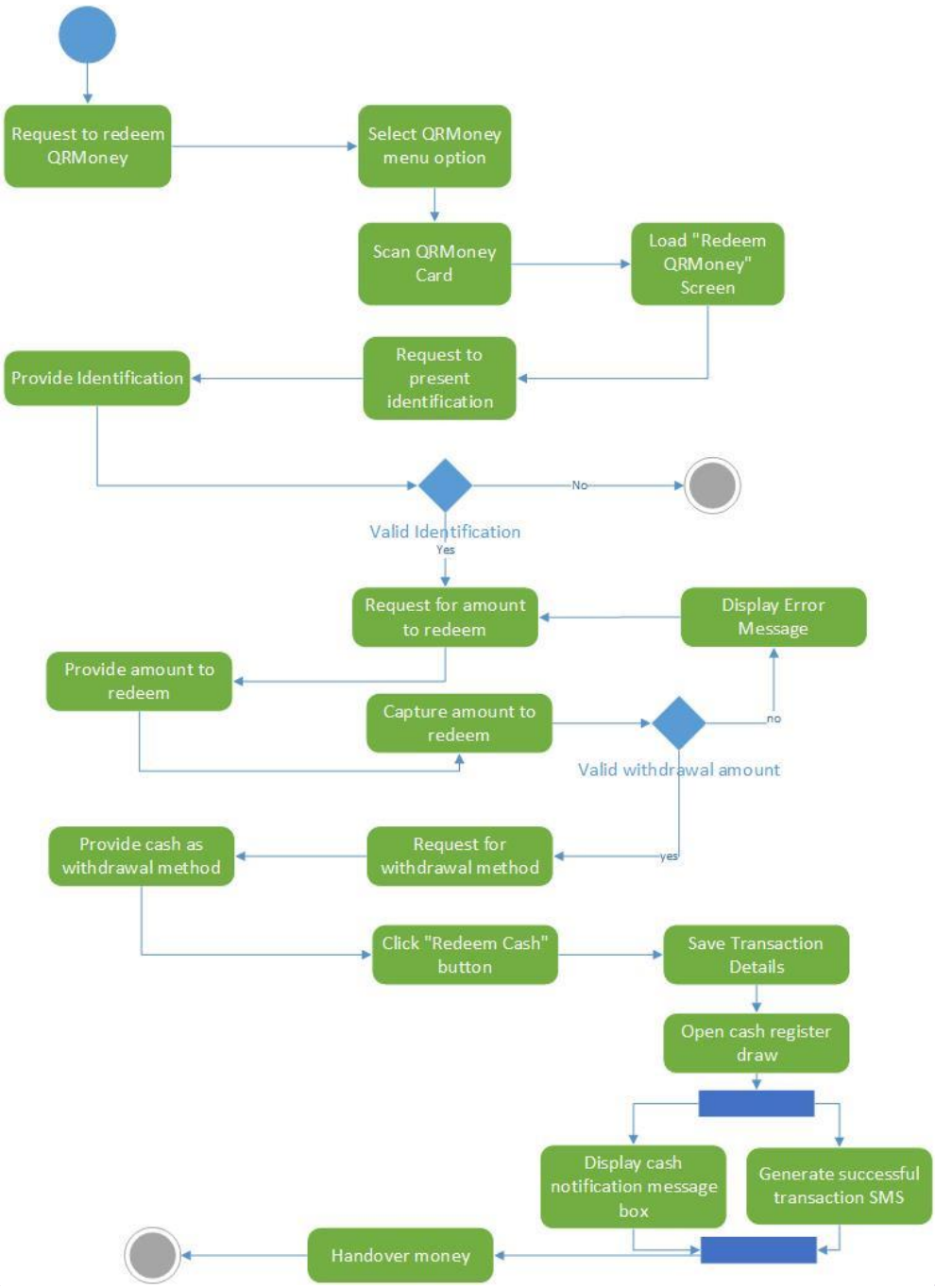


Commented [DMH1]: Last flow through API not GUI

Question 2 [20 marks]

Draw the **UML Activity Diagram** for use case 5.1 Redeem QRMoney. **Only draw the typical course of events.**

Activity – 9 (1/2 mark each)
Start node – 1
End node – 2 (1 mark each)
Decision node – 2 (1 mark each)
Join node – 1
Data flows – max. 5 (1/2 mark each – stop marking once reach 10 correct data flows)



Commented [DMH2]: First money withdrawn then sent SMS

Question 3 [8 marks]

Before a decision is made on whether a project should continue or not, a feasibility study needs to be conducted. Explain the four types of feasibility tests, with appropriate examples, that need to be done.

1 mark per feasibility type, 1 mark per explanation

Technical feasibility - The extent to which the system can be successfully designed, developed and installed by the IT group "Can we build it?"

- *Familiarity with the application*
- *Familiarity with technology*
- *Project size*
- *Compatibility*

Organizational feasibility - A measure of how well a solution will work and be accepted in an organization. Is the project strategically aligned with the business? If solution makes technical sense but conflicts with the way the organisation does business, it is not feasible. Includes issues such as:

- *Personnel policies*
- *Redrawing job specifications*
- *Threats to industrial relations*
- *Expected skills requirements*
- *Motivation*
- *Support from stakeholders, management and users*

Schedule feasibility - A measure of how reasonable the project timetable is

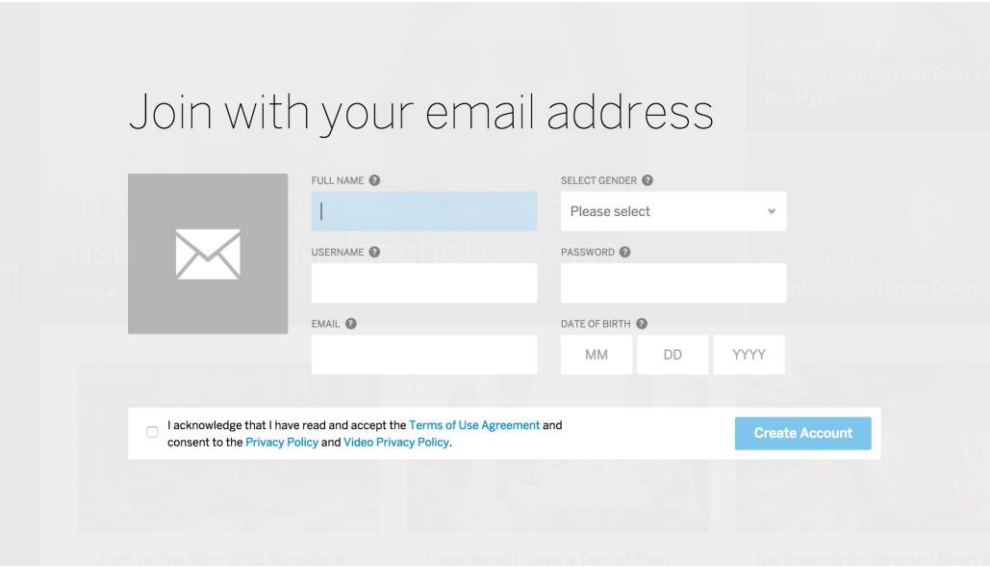
- *A project will fail if it takes too long to develop*
- *It entails the estimation of how long it will take to develop the system.*
- *Project management plays a crucial role*

Economic feasibility - A measure of the cost-effectiveness of a project or solution. Any project has economic costs and economic benefits. Has three strands

- *Benefits must justify costs*
- *Project must be 'best option'*
- *Project competes with other business areas for funds*

Question 4 [5 marks]

Analyse the following screen design by identifying five good or bad design elements.



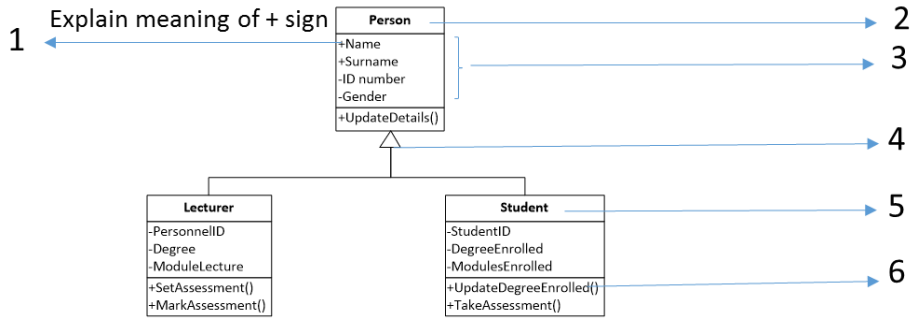
Memo: (Any 5)

Student answers will differ.
Suggested good elements:
Clean design, no clutter
All labels left aligned, above text boxes
Dropt down for options
Indicating the format in which the date of birth needs to be completed.
Boxing of the ToUA makes it stand out
Help function available

Suggested bad elements
Create account does not look like a button

Question 5 [6 marks]

1. Name the following elements associated with a class diagram.



1 Attribute is public – not hidden from any other objects
2 Super/Parent class Person / Class name compartment
3 Attributes / also attributes compartment
4 Generalisation symbol / Inheritance
5 Sub/Child class
6 Operation / Method

Question 6 [6 marks]

The output design lesson introduced two different types of reports. Identify and explain two types of reports and provide an appropriate example for each **from the case study**.

Answer:
Internal report ✓: Internal to the company, ✓Monthly report to Mr Trepeneur to detail number of transactions etc.✓
External report✓: Going outside the company ✓ To the SP detailing the number of transactions, money available and money redeemed. ✓